HMI-01 LABVIEW SCADA FUNCTIONAL ARCHITECTURE

PLC REQUEST / RESPONSE / ALARM / TREND / EVENT LOGGING CONCEPT

A. OPERATOR / HMI FRONT PANEL

MAIN OVERVIEW
AUTO OPERATION
MANUAL / SETUP
ALARM / HISTORY
TREND / PRODUCTION
SETTINGS / USER ACCESS
COMMUNICATION / DIAGNOSTIC

- OPERATOR INPUT
- MODE SELECT
- SETPOINT ENTRY
- AUTO START REQUEST
- ALARM ACKNOWLEDGE
- ALARM RESET REQUEST
- COUNT RESET REQUEST
- USER LOGIN / LOGOUT

SEE DWG 016 / SHEET 1 OF 3, SHEET 2 OF 3, AND SHEET 3 OF 3.

PLC STATUS / SEQUENCE / COMMUNICATION DISPLAY

USER COMMAND / INPUT REQUEST

TREND / PRODUCTION STATUS DISPLAY

B. HMI-01 LABVIEW APPLICATION

B1. USER INTERFACE EVENT HANDLER

MODE SELECT REQUEST
SETPOINT WRITE REQUEST
AUTO / RESET / ACK REQUEST
COUNT RESET REQUEST
USER ACCESS CHECK

ONLY AUTHORIZED AND VALIDATED USER REQUESTS SHALL BE FORWARDED TO PLC-01.

VALIDATED HMI COMMAND REQUEST

B2. PLC COMMAND / RESPONSE HANDLER

MODBUS TCP CLIENT
WRITE REQUEST REGISTER / BIT
READ STATUS / SEQUENCE / ALARM DATA
READ SETPOINT / PRODUCTION DATA
COMMUNICATION WATCHDOG

HMI-01 SHALL WRITE PLC REQUESTS ONLY.
PLC-01 SHALL VALIDATE MODE, STATE, RANGE, FAULT, AND SAFETY CONDITIONS BEFORE EXECUTING COMMANDS.

B3. MCU COMMUNICATION HANDLER

MODBUS RTU MASTER
READ MCU STATUS / PROCESS DATA
READ PROXIMITY / VIBRATION DATA
MCU COMMUNICATION WATCHDOG
MODBUS RTU / SERIAL

PLC ALARM / FAULT / COMMUNICATION STATUS

MCU COMMUNICATION FAULT / WATCHDOG STATUS

ACTIVE ALARM / ALARM HISTORY DISPLAY

B4. ALARM / EVENT PROCESSOR

ACTIVE ALARM DETECTION
ALARM ACKNOWLEDGE STATUS
ALARM RESET STATUS
ALARM HISTORY RECORD
COMMUNICATION FAULT EVENT

DATA LOGGING / FILE WRITE FAULT

PLC STATUS / SEQUENCE / PRODUCTION DATA

B5. TREND / PRODUCTION DATA PROCESSOR

PROCESS / VIBRATION TREND
PROXIMITY TREND
MOTOR SPEED TREND
TEST POSITION TREND
CYCLE / GOOD / REJECT COUNT
CYCLE TIME CALCULATION

ALM-001 / ALM-101~ALM-108 SEE DWG 012.

MCU PROCESS DATA / SENSOR STATUS

B6. HISTORICAL DATA ACCESS / CSV EXPORT HANDLER

READ ALARM EVENT HISTORY
READ TREND DATA FILE
READ PRODUCTION RESULT LOG
APPLY HISTORY FILTER
GENERATE FILTERED CSV EXPORT
CSV EXPORT STATUS
CSV EXPORT
LOGGING STATUS MONITOR
HMI-01 SYSTEM CLOCK / TIME STAMP SOURCE
TIME SYNCHRONIZATION STATUS
FILE WRITE ERROR DETECTION
DATA STORAGE FAULT EVENT

C. PLC-01 MODBUS TCP SERVER / PLC CONTROL SYSTEM

HMI REQUEST MEMORY
PLC COMMAND VALIDATION
SEQUENCE STATE MACHINE
ALARM / FAULT LOGIC
SETPOINT RANGE CHECK
PRODUCTION COUNT DATA

HMI COMMUNICATION LOSS SHALL NOT BYPASS OR DISABLE HARDWIRED SAFETY.

SEE DWG 011 / DWG 012 / DWG 013 / DWG 015.

HMI REQUEST WRITE

PLC STATUS / ALARM / DATA READ

D. MCU-01 PROCESS SENSOR / MODBUS RTU DEVICE

FIELD SENSOR DATA PROVIDER
VIBRATION DATA
PROXIMITY DATA
MCU STATUS
COMMUNICATION STATUS

MCU-01 PROVIDES PROCESS DATA ONLY.
PLC-01 REMAINS THE PRIMARY CONTROL AUTHORITY.
MCU COMMUNICATION FAULT SHALL BE RECORDED AS HMI ALARM EVENT.

MODBUS RTU POLL / READ REQUEST

PROCESS DATA / STATUS RESPONSE

E. HMI DATA STORAGE

- ALARM / EVENT HISTORY — ALARM EVENT RECORD
- TREND DATA FILE — TREND / PROCESS DATA RECORD
- PRODUCTION RESULT LOG — PRODUCTION RESULT RECORD
- OPERATOR ACTION LOG — USER ACTION AUDIT RECORD
- CSV EXPORT FILE — FILTERED CSV EXPORT

HISTORICAL DATA READ BUS

ALARM / TREND / PRODUCTION LOG DATA

HMI-01 SHALL OPERATE AS A SUPERVISORY SCADA INTERFACE.
ALL PROCESS CONTROL COMMANDS SHALL BE EXECUTED ONLY AFTER PLC-01 VALIDATION.

HMI-01 SHALL NOT DIRECTLY CONTROL PLC OUTPUTS P20~P29, OVERRIDE PLC INTERLOCKS, OR REPLACE HARDWIRED SAFETY FUNCTION.

HMI TAG / REQUEST / RESPONSE / LOGGING MAP:
SEE DWG 015 / SHEET 1 OF 3, SHEET 2 OF 3, AND SHEET 3 OF 3.

ALL ALARM, TREND, PRODUCTION, AND OPERATOR ACTION RECORDS SHALL BE TIME-STAMPED BY THE HMI-01 SYSTEM CLOCK.

TIME SYNCHRONIZATION FAULT SHALL BE INDICATED ON HMI-01 SETTINGS SCREEN.

[CONNECTION LEGEND]

- ARROWHEAD — SIGNAL / DATA FLOW DIRECTION
- LINE CROSSING WITHOUT FILLED DOT — NO FUNCTIONAL CONNECTION
- FILLED DOT OR T-JUNCTION — FUNCTIONAL BRANCH / CONNECTION

[FLOW LEGEND]

- YELLOW ARROW — HMI COMMAND / REQUEST / POLL FLOW
- GREEN ARROW — STATUS / PROCESS DATA / LOG DATA FLOW
- RED ARROW — ALARM / FAULT / SAFETY RELATED EVENT FLOW

NO.	DATE	REVISION
A	2026-09-19	INITIAL RELEASE

DWG TITLE

EDS Wafer Map Linked Package
Final Test and Multi-Bin Sorting Cell

LABVIEW SCADA FUNCTIONAL ARCHITECTURE / COMMUNICATION / DATA

DRAWN BY	DATE
J. H. KO	2026-09-19

DWG NO: 017

SHEET NO: 1 OF 1